MOHAMMAD FARAJI
Machine Learning Researcher

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Representation Learning Lab (222), Computer Engineering Department,University of Kurdistan, Sanandaj, Iran



BRIEFLY

I am a Machine Learning Researcher at the University of Kurdistan, where my research focuses on representation learning, feature selection, and multi-label learning. My research interests lie in developing fundamental methodologies for supervised, semi-supervised, and unsupervised machine learning, with particular emphasis on representation learning models and their applications in complex learning scenarios. I received my Master's degree in Artificial Intelligence from the Department of Computer Engineering at the University of Kurdistan in 2023. My Master's research was supervised by Dr. Fardin Akhlaghian and advised by Amjad Seyedi. During my graduate studies, I focused primarily on matrix factorization and low-rank approximation techniques, particularly for supervised multi-label feature selection and multi-label learning problems. Prior to pursuing my graduate studies, I obtained Associate and Bachelor's degrees in Software Engineering, which provided me with a strong foundation in computational methods and algorithmic development.

RESEARCH INTERESTS

- **Machine Learning, Deep Learning, Representation Learning**
- Multi-Label Learning, Feature Selection, Optimization
- Numerical Linear Algebra, Low-Rank Approximation, Matrix Factorization

EXPERIENCE

ML Researcher
Research

University of Kurdistan, SANANDAJ, IRAN, (Sep 2021 – Present)
- Conducting research in representation learning, feature selection, and multi-label learning.
- Developing machine learning models and optimization algorithms using Python and MATLAB.

Research Service
National Service

University of Kurdistan, SANANDAJ, IRAN, (Oct 2024 – Sep 2026)
- Managing and organizing academic library resources at the Central Library of the University of Kurdistan.
- Developing a research proposal on improving efficiency using artificial intelligence methods (proposal defense stage).

EDUCATION

Master

Artificial Intelligence, UNIVERSITY OF KURDISTAN, SANANDAJ, IRAN, (Sep 2020 – Sep 2023)
- Thesis title : Multi-Label Feature Selection with Global and Local Label correlation.
- Superviser : Dr. Fardin Akhlaghian.
- Courses : machine learning, statistical pattern recognition, neural networks, advanced artificial intelligence, computer vision, digital image processing, complex Networks, and special subjects.

Bachelor

Sofware Engineering, SHAMSIPOUR TECHNICAL COLLEGE, TEHRAN, IRAN, (Sep 2017 – Jul 2019)
- Project title : Student Management system.

Associate

Computer Software, YAZDANPANAH TECHNICAL COLLEGE, SANANDAJ, IRAN, (Sep 2015 – Sep 2017)
- Supplementary courses in computer science and software engineering.

PUBLICATION

In-Prep

Imbalance-Aware Consensus Learning and Adaptive Rescaling for Multi-View Multi-Label Feature Selection
M. Faraji, S. A. Seyedi, and F. Akhlaghian.

Submitted

Anchor Mediated Hierarchical Feature Selection through Shared Structure Learning for Multi Label Data
R. Mahmoodi, **M. Faraji**, and F. Akhlaghian.
Information Fusion.

2026 | **Robust log-based multi-label feature selection with dynamic label correlation and relevance-redundancy optimization**
M. Faraji, S. A. Seyedi, and F. Akhlaghian.
Knowledge Based-Systems, volume 342, 2026, pp.115825.

2024 | **Mutli-Label Feature Selection with Global and Local Label Correlation**
M. Faraji, S. A. Seyedi, F. Akhlaghian, and R. Mahmoodi.
Expert Systems with Applications, volume 246, 2024, pp.110179.

COMPUTER SKILLS

Operation Systems	Microsoft Windows and Linux
Word processing & Presentation	Office suites and LaTeX
Vector and raster softwares	Adobe Illustrator and Abode XD
Development Tools	Pycharm, Jupyter Notebook, Colab, and Visual Studio
Web design	HTML, CSS, ASP.NET, and JavaScript

PROGRAMMING LANGUAGES

2021 – present	**Python**, PyTorch, NumPy, Pandas, and scikit-learn
2022 – present	**MATLAB**, linear algebra and visualization
2015 – 2017	**C++ | C#**, Software Engineering and Web development
2013 – 2015	**Basic | Visual Basic**, Software Engineering

PROJECT

Patter Recognition	implemented several embedded feature selection algorithms, including methods based on label correlation analysis and robust optimization.
Image Processing	Implemented image enhancement, segmentation, feature extraction, and object recognition algorithms using MATLAB.
Neural Networks	Implemented associative memory and optimization algorithms based on Hopfield neural networks and analyzed convergence behavior.
Machine Learning	Implemented classification, clustering, dimensionality reduction, and ensemble learning algorithms for benchmark datasets.

REFERENCES

Fardin Akhlaghian, *Associate Professor*
Department of Computer Engineering, University of Kurdistan, Sanandaj, Iran
Relationship : Master's Thesis Supervisor & Research Supervisor
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Amjad Seyedi, *Doctoral Researcher*
Department of Mathematics and Operational Research, Polytechnic Faculty of Mons, University of Mons, Belgium
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Parham Moradi, *Senior Lecturer*
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Alireza Abdollahpouri, *Associate Professor*
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